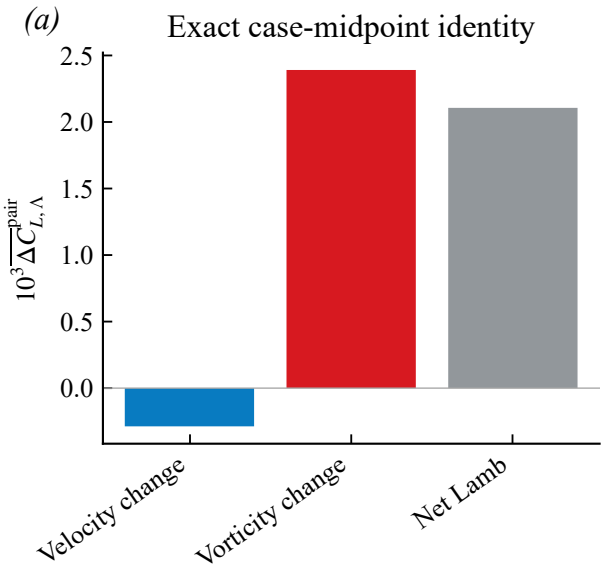
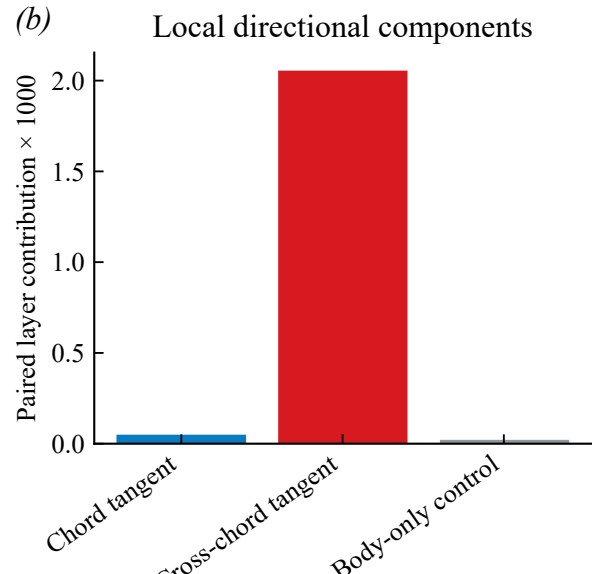
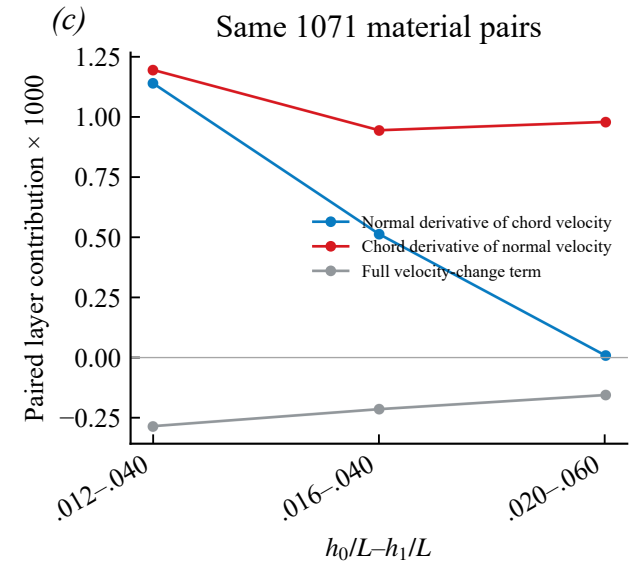

(a)
Exact case-midpoint identity
$10^3\overline{\Delta C}^{\mathrm{pair}}_{L,\Lambda}$
2.5
2.0
1.5
1.0
0.5
0.0
Velocity change
Vorticity change
Net Lamb


(b)
Local directional components
Paired layer contribution × 1000
2.0
1.5
1.0
0.5
0.0
Chord tangent
Cross-chord tangent
Body-only control


(c)
Same 1071 material pairs
Paired layer contribution × 1000
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
Normal derivative of chord velocity
Chord derivative of normal velocity
Full velocity-change term
.012–.040
.016–.040
.020–.060
h_0/L–h_1/L